# PlanRadar Form Builder & Validation

*Test Plan, Automation Report & QA Technical Assessment Findings*

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## 1. Executive Summary & Strategy

This Test Plan outlines the testing strategy, scenario matrix, and automated test implementation for the PlanRadar Form Builder and Ticket Creation modules. The quality assurance strategy combines manual exploratory/system verification with an automated End-to-End (E2E) regression pipeline to ensure robust form creation, accurate field property propagation, strict role-based permission enforcement, default value pre-population, and mandatory field validation.

## 2. Scope of Testing & Assessment Coverage

### Detailed Functional Scope

**1. Form Creation & Field Configurations:** Creation of dynamic form templates using diverse field types (Title, Short Text, List/Dropdown, Date, Checkbox, Number, User, Attachment, Checklist, Separators, Groups) and form-level feature toggles (Attachments, Comments, Plan Position, Geographic Location, Recurring Activity, Locking, Signatures, Numbering Sequence, Approval Workflows).

**2. Role-Based Field Permissions:** Configuring fine-grained role permissions ('Visible To' and 'Editable By') and verifying access control enforcement across Administrator, Subcontractor, and custom project roles during ticket creation.

**3. Field Default Properties:** Defining global property default values in the form builder and validating automatic population, editing, and clearing behavior inside the ticket creation modal.

**4. Mandatory Field Enforcement:** Enabling mandatory field rules, validating visual indicators ('*') in both Form Canvas and Ticket Preview Modal.

**5. E2E Automation Pipeline:** Full E2E automated execution using Java 21, Selenium WebDriver, and TestNG to automate authentication, form creation, mandatory configuration, ticket preview modal verification and clean navigation exit.

# 3. Detailed Test Cases Matrix

## Scenario 1: Form Creation, Field Types & Configurations

| TC ID | Test Case Title | Test Steps | Expected Result | Priority |
|---|---|---|---|---|
| TC1.1 | Create Form with Multiple Field Types | 1. Open Form Builder.<br>2. Add Short Text, List, Date, Title, and Checkbox fields onto canvas.<br>3. Save form. | Form is created successfully and all field types render accurately on the canvas. | High |
| TC1.2 | Configure Form-Level Feature Settings | 1. Open Form Settings sidebar.<br>2. Enable Attachments, Comments, Plan position, Signatures, and Numbering sequence.<br>3. Save form. | All enabled form-level features are saved and applied to ticket creation. | High |
| TC1.3 | Configure List Field Dropdown Options | 1. Select List field on canvas.<br>2. Add custom dropdown options ('Option A', 'Option B').<br>3. Save form. | Dropdown options save correctly and appear during ticket creation. | Medium |
| TC1.4 | Canvas Field Selection & Renaming | 1. Click Short Text field on canvas.<br>2. Change Field Name to 'QC Inspection Note'. | Field label updates immediately across property panel and canvas card. | Medium |
| TC1.5 | Configure Form Ticket Permissions | 1. Open Form Settings.<br>2. Set Ticket Permissions to 'All roles can view and create'. | Permission rule enables form access across all user roles. | High |
| TC1.6 | Reorder Canvas Fields via Drag and Drop | 1. Add 3 fields to canvas.<br>2. Drag fields to reorder using drag handles.<br>3. Save form. | Field layout on canvas and in ticket creation modal reflects new order. | Medium |
| TC1.7 | Duplicate & Delete Canvas Fields | 1. Click Duplicate on field card.<br>2. Click Delete icon on duplicate field. | Field duplicates with identical properties and is removed cleanly on delete. | Low |
| TC1.8 | Forms Management List Search & Filter | 1. Navigate to Forms list page.<br>2. Search form name in search bar. | Forms list accurately filters and displays the target form template. | Medium |

## Scenario 2: Role-Based Field Permissions Enforcement

| TC ID | Test Case Title | Test Steps | Expected Result | Priority |
|---|---|---|---|---|
| TC2.1 | Configure Field View & Edit Permissions | 1. Select field on canvas.<br>2. Disable 'All roles can view and edit'.<br>3. Set Visible To = Administrator, Subcontractor.<br>4. Set Editable By = Administrator only. | Field permission rules save successfully in property panel. | High |
| TC2.2 | Verify Field Access as Administrator Role | 1. Log in as Administrator.<br>2. Open Create Ticket modal with form. | Field is fully visible and editable by the Administrator. | High |
| TC2.3 | Verify Field Access as Subcontractor Role | 1. Log in as Subcontractor.<br>2. Open Create Ticket modal with form. | Field is visible to Subcontractor but strictly read-only (disabled input). | High |
| TC2.4 | Verify Field Access for Restricted Role | 1. Log in as user role not listed in Visible To.<br>2. Open Create Ticket modal. | Field is completely hidden from the ticket creation modal. | High |
| TC2.5 | Real-Time Permission Change Enforcement | 1. Modify field permissions in Form Builder.<br>2. Log in as affected role and open ticket modal. | Updated permissions take effect immediately on new ticket sessions. | Medium |

## Scenario 3: Field Default Values Functionality

| TC ID | Test Case Title | Test Steps | Expected Result | Priority |
|---|---|---|---|---|
| TC3.1 | Set Default Value in Form Builder | 1. Select Short Text field.<br>2. In Global Properties, set Default Value = 'test'.<br>3. Save form. | Default value 'test' is saved and bound to the Short Text field. | High |
| TC3.2 | Verify Auto-Population during Ticket Creation | 1. Open Tickets module.<br>2. Click Create Ticket and select form. | Short Text field is automatically pre-filled with 'test'. | High |
| TC3.3 | Override Pre-filled Default Value | 1. Open Create Ticket modal with pre-filled value 'test'.<br>2. Replace with 'Custom Entry'.<br>3. Save ticket. | Ticket saves successfully displaying 'Custom Entry' in ticket details. | High |

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| TC3.4 | Clear Pre-filled Default Value and Save | 1. Open Create Ticket modal with default value pre-filled.<br>2. Clear default value completely.<br>3. Click Save. | Field should clear and save with empty value (or throw required error). ACTUAL: Reverts to pre-filled default value upon save. | Medium |
| TC3.5 | Default Values for List and Date Fields | 1. Set default choice for List field and default date.<br>2. Open Create Ticket modal. | Pre-configured dropdown option and date load automatically. | Medium |

## Scenario 4: Mandatory Field Enforcement & Validation

| TC ID | Test Case Title | Test Steps | Expected Result | Priority / Status |
|---|---|---|---|---|
| TC4.1 | Enable Mandatory Flag on Form Canvas | 1. Select Short Text field on canvas.<br>2. Check Mandatory toggle in field properties. | Field label renders with mandatory asterisk '* Short Text' on canvas. | Medium |
| TC4.2 | Validation Error on Blank Mandatory Field | 1. Open Create Ticket modal.<br>2. Clear mandatory Short Text field.<br>3. Click Save. | Submission is blocked. Field highlights red with banner 'You have not filled in all the required fields.' | High |
| TC4.3 | Satisfy Mandatory Field & Submit Ticket | 1. Enter valid text into mandatory Short Text field.<br>2. Click Save. | Validation error clears and ticket is created successfully. | High |
| TC4.4 | Multiple Mandatory Fields Enforcement | 1. Set Short Text, List, and Date fields as Mandatory.<br>2. Attempt ticket save with 1 empty mandatory field. | Form highlights specific missing required field and blocks ticket creation. | High |

## 4. Defect Report & Bug Documentation

During the manual execution of Scenario 3 (TC3.4), a functional defect was identified regarding default value clearing persistence:

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| **Defect ID** | BUG-TC3.4 |
| **Test Case ID** | TC3.4 (Clear Pre-filled Default Value) |
| **Severity / Priority** | Medium / High |
| **Summary** | Clearing a pre-filled default value in a ticket field does not clear the field upon saving; the system retains or reverts to the default value. |
| **Environment** | Web Application (Chrome v130+, Windows 11, PlanRadar Demo Environment) |
| **Steps to Reproduce** | 1. Open Form Builder and set a default value (e.g. 20/10/2026) for a date field.<br>2. Save form and navigate to Tickets module.<br>3. Click '+ Create Ticket' and select the form.<br>4. Observe field auto-populated with “20/10/2026”.<br>5. Clear the text/date from the field completely.<br>6. Click 'Save'. |
| **Expected Result** | The ticket saves with an empty field value (or displays a required field error if configured as mandatory). |
| **Actual Result** | The cleared field value is ignored upon saving; the system reverts back to the original pre-filled default value (“20/10/2026”). |
| **Attachement** | BUG-TC3.4-Attachement |

## 5. Recommendations for Product Improvements & Enhancements

**Form Builder Usability Enhancements:**
Improvements that will enhance user experience and eliminate UI friction in the Form Builder feature:

**1. Single-Click Form Saving:** Currently, creating or updating a form requires clicking 'Save' twice, or prompts an 'Unsaved Changes' confirmation modal even after save is clicked. Streamlining the save mechanism so changes persist immediately upon the first click will eliminate user confusion.

**2. Action Icon Tooltips / Helptips:** Field action icons on the canvas (Edit, Duplicate, Delete) currently lack explanatory hover labels. Adding action helptips/tooltips on hover (e.g. 'Edit field settings', 'Duplicate field', 'Delete field') will significantly improve accessibility and feature discoverability.

## 6. Automation Testing Process: Observations & Challenges

Key observations and UI challenges encountered during Selenium automation setup:

**1. Cookie Consent Banner:** The initial cookie consent banner overlays the UI on launch and blocks navigation clicks. It must be explicitly accepted before running test steps.

**2. Time zone Handling:** Date field inputs and assertions can vary based on system and browser time zone settings. Ensuring standardized date formats across environments prevents date-picker mismatches.

## 7. Automation Framework Enhancements & Script Strategy

**Mandatory Field Validation Enhancement:**
Script Enhancement for Ticket Creation Mandatory Assertion:

The current automated script verifies mandatory fields using Preview Mode by asserting the presence of the required asterisk inside * Short Text . The framework is enhanced to perform real-time mandatory field assertions directly inside the Create Ticket modal by attempting ticket submission with an empty field. This validates: (1) Submission blocking upon clicking Save, (2) Red border field error highlighting, and (3) Verification of the inline validation message 'You have not filled in all the required fields.' / 'Error: Cannot be empty'.

## 8. Suite Execution & Reporting

**GitHub Repository:** Github-repo

**IntelliJ IDEA TestNG Runner:** Right-click testng.xml in project root directory and select Run '.../testng.xml'.